

Lucas COEURET

Born on 20 September 1996, in Rouen (76), France

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	<i>Research</i>	<i>Teaching</i>
	IECL (Site de Nancy)	IUT Nancy-Charlemagne
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	Bureau 317	Bureau 114

Academic position

— 2025-Present *Assistant Professor (Maître de conférences stagiaire, Section CNU 26)*

University: Université de Lorraine

Research laboratory: Institut Elie Cartan de Lorraine (IECL), UMR CNRS 7502

Teaching unit: IUT Nancy-Charlemagne

— 2024-2025 *Postdoctoral researcher (Assegnista di ricerca)*

University: Università degli studi di Padova

Research laboratory: Dipartimento di Matematica Tullio Levi-Civita

Supervisors: Fabio ANCONA, Roberta BIANCHINI, Laura CARAVENNA, Elio MARCONI

Project: PRIN Project 2022 PNRR (No. 2022XJ9SX) entitled “Heterogeneity on the road - Modeling, analysis, control”

Academic background

— 2021-2024 *PhD in Applied Mathematics*

University: Université Toulouse 3 Paul Sabatier

Research laboratory: Institut de Mathématiques de Toulouse (IMT), UMR CNRS 5219

Advisors: Jean-François COULOMBEL and Grégory FAYE

Title: Stability of discrete shock profiles for systems of conservation laws

Defense: 12 July 2024

Referees: Hermen Jan HUPKES, Pauline LAFITTE

Jury: Claire CHAINAIS-HILLAIRET (*President of the jury*), Frédéric ROUSSET

Financing: “Contrat Doctoral Spécifique pour Normalien” of the ENS Paris-Saclay

— 2017-2021 *Élève normalien at ENS Paris-Saclay, Cachan*

- **2020-2021** M2 Mathématiques de la Modélisation at Sorbonne Université with highest honors
- **2020** Admitted 29th at the competitive exam of the Agrégation of Mathematics (option B scientific calculus)
- **2019-2020** M2FESup in Mathematics option B, ENS Paris-Saclay, preparation for the Agrégation of Mathematics with highest honors
- **2018-2019** M1 Hadamard, ENS Paris-Saclay (with courses at Université de Paris-Sud and École Polytechnique) with high honors
- **2017-2018** L3 in Mathematics from ENS Paris-Saclay in collaboration with Université Paris-Diderot with high honors
- **2017** Admitted to ENS Paris-Saclay (Competitive exam)

— 2014-2017 *MPSI and MP* at Lycée Louis-le-Grand, Paris*

— 2014 Scientific Baccalauréat (section Abibac) with highest honors at Lycée Gustave Flaubert, Rouen

Research interests

- Finite difference schemes for the transport equation: Stability, large-time behavior, boundary conditions, GKS theory
- Conservative finite difference schemes for hyperbolic systems of conservation laws: stability of discrete shock profiles via spatial dynamics technique and Green's function analysis
- Traffic flow models

List of (pre)publications

— Preprints

— Publications

- L. COEURET — **Nonlinear orbital stability of stationary discrete shock profiles for scalar conservation laws**
Accepted at *Annales Mathématiques Blaise Pascal*, (2024), pp. 1-49
HAL: <https://hal.science/hal-04712769> arXiv: <https://arxiv.org/abs/2409.18930>
- L. COEURET — **Linear stability of discrete shock profiles for systems of conservation laws**
Journal of Hyperbolic Differential Equations, (2025), Vol. 22, No. 02, pp. 261-414
Journal: <https://doi.org/10.1142/S0219891625500092> arXiv: <https://arxiv.org/abs/2311.02507>
HAL: <https://hal.science/hal-04270648>
- L. COEURET — **Local limit theorem for complex valued sequences**
Asymptotic Analysis, (2025), 142 (2), pp. 379-431
DOI: <https://doi.org/10.1177/09217134241308379> arXiv: <https://arxiv.org/abs/2201.01514>
HAL: <https://hal.science/hal-03463375>
- L. COEURET — **Tamed stability of finite difference schemes for the transport equation on the half-line**
Mathematics of Computation, (2024), Volume 93, Number 347, pp. 1097-1151
DOI: <https://doi.org/10.1090/mcom/3901> arXiv: <https://arxiv.org/abs/2304.02612>
HAL: <https://hal.science/hal-04059973>

— Proceedings

- L. COEURET — **Large time behavior of finite difference schemes for the transport equation**
Hyperbolic Problems: Theory, Numerics, Applications. Volume II. HYP 2022. SEMA SIMAI Springer Series, Vol 35., (2024), pp. 63-71
DOI: https://doi.org/10.1007/978-3-031-55264-9_6 HAL: <https://hal.science/hal-04191971>

— Thesis

- **Stability of discrete shock profiles for systems of conservation laws**
Defended on 12 July 2024, at the Institut de Mathématiques de Toulouse, Université Toulouse III Paul Sabatier
HAL: <https://theses.hal.science/tel-04652121>

Seminars, Conferences and Scientific Schools

— Talks at seminars

- Séminaire d'analyse numérique du LmB, 23 April 2026, Besançon (France)
- Séminaire EDP et Applications de l'Institut Élie Cartan de Lorraine, 23 January 2026, Metz (France)
- Groupe de travail EDP de l'Institut Élie Cartan de Lorraine, 20 January 2026, Nancy (France)
- Séminaire EDP et Applications de l'Institut Élie Cartan de Lorraine, 25 March 2025, Nancy (France)
- Seminari Generali dell'IAC, 8 October 2024, Rome (Italia)
- Séminaire EMA, 23 November 2023, Calais (France)
- Séminaire d'Analyse Numérique de l'IRMAR, 16 November 2023, Rennes (France)

— Participation to conferences with talk

- SIAM Conference on Nonlinear Waves and Coherent Structures (NWCS26), 26-29 May 2026, Montréal (Canada)
Presentation of a 20 minutes talk
- Workshop ANR COSS, 5-6 Mars 2026, Tours (France)
Presentation of a 45 minutes talk
- HyPNuT : Hyperbolic Prolems - Numerics and Theory, 5-7 November 2025, Amiens (France)
Presentation of a 40 minutes talk
- Workshop: Traffico e Leggi di Conservazione, 5 June 2025, Rome (Italy)
Presentation of a 40 minutes talk
- Conference on Mathematics of Wave Phenomena, 24-28 February 2025, Karlsruhe (Germany)
Presentation of a 25 minutes talk
- Journées Jeunes EDPistes 2025, 8-10 January 2025, Nice (France)
Presentation of a 25 minutes talk
- Equadiff 2024, 10-14 June 2024, Karlstad (Sweden)
Presentation of a 30 minutes talk
- CANUM (Congrès d'Analyse Numérique) 2024, 27-31 May 2024, Île de Ré (France)
Presentation of a 30 minutes talk
- Congrès des Jeunes Chercheurs en Mathématiques et Applications 2023, 25-27 September 2023, Gif-sur-Yvette (France)
Presentation of a 20 minutes talk
- Numhyp23 : Numerical Methods for Hyperbolic Problems, 26-30 June 2023, Bordeaux (France)
Presentation of a 20 minutes talk
- HYP 2022: XVIII International Conference on Hyperbolic Problems: Theory, Numerics, and Applications, 20-24 June 2022, Málaga (Spain)
Presentation of a 20 minutes talk

— Participation to conferences without talk

- Journées ANR HEAD, 13-16 April 2026, Toulouse (France)
- Mécanique des fluides dans le Grand Est , 27-28 January 2026, Strasbourg (France)
- Summer School on “Population Dynamics: From fundamental to applied science”, 16-27 June 2025, Grasse (France)
- 13th Meeting on Nonlinear Evolution PDEs, Fluid Dynamics and Transport Equations, 17-21 February 2025, Trieste (Italia)
Participation to the poster session

- **Summer school in fluid dynamics and nonlinear PDEs**, 9-13 September 2024, Padova (Italia)
- **Workshop on spatial dynamics and related approaches**, 5-7 September 2022, Stuttgart (Germany)
Participation to the poster session

Teaching

IUT Nancy-Charlemagne, BUT Informatique

2025-2026 160HETD

Université Toulouse III - Paul Sabatier, Mathematics department

2021-2024 64HETD each year, $L1$ and $L2$,

- *Course*: Introduction to real analysis
- *Teaching assistant (TD)*: Functions and Calculus 1, Introduction to real analysis, Numerical Methods
- *Practical sessions (TP)*: Introduction to real analysis, Numerical Methods, Linear Algebra

Organizing responsibilities

- **02/2024 - 09/2024** Representative of the PhD Students and Postdoctoral Researchers on the Laboratory Council at the Institut de Mathématiques de Toulouse
- **09/2023 - 09/2024** Organizer of the PDE student seminar at the Institut de Mathématiques de Toulouse

Internships

- **2021 - April / July** Scientific traineeship at Institut de Mathématiques de Toulouse, Université Toulouse 3 Paul Sabatier under the guidance of Jean-François COULOMBEL and Grégory FAYE (4 Months)
- **2019 - April / July** Scientific traineeship at Institut de Mathématiques de Toulouse, Université Toulouse 3 Paul Sabatier: Laurent Operators and Stability of Numerical Schemes, under the guidance of Jean-François COULOMBEL (4 Months)
- **2018** Scientific traineeship at CMLA, ENS Paris-Saclay: Theoretical and numerical study of an inequality in fluid mechanics with Yasmine EL-KAOUNI, under the guidance of Laure QUIVY (3 Months)
- **2015 - July / August** Linguistic traineeship at Humboldt Universität, Berlin

Languages and Various Information

- German (Abitur with a note of 1.8, Level C1 certification of the Humboldt Universität, Level B2 certification of the Goethe Institut)
- English (Level C2 certification at the Cambridge Advanced Exam)
- Programming skills in Python and L^AT_EX